6.2 Solution: (a) Let $\mathbf{x}_0'$ be a solution to the vector equation

$$\mathbf{f}(\mathbf{x}') = \mathbf{x}' - \mathbf{r}\left(t - \frac{|\mathbf{x}-\mathbf{x}'|}{c}\right) = 0,$$

then the Dirac delta function $\delta(\mathbf{f}(\mathbf{x}'))$ can be expressed as

$$\delta(\mathbf{f}(\mathbf{x}')) = \frac{\delta(\mathbf{x}'-\mathbf{x}_0')}{|\det[J\mathbf{f}(\mathbf{x}_0')]|},$$

where $J\mathbf{f}(\mathbf{x}_0')$ is the Jacobian of $\mathbf{f}$ at $\mathbf{x}' = \mathbf{x}_0'$. It is straightforward to show that

$$J\mathbf{f}(\mathbf{x}_0') = \begin{pmatrix} 1 - v_x n_x/c & -v_x n_y/c & -v_x n_z/c \\ -v_y n_x/c & 1 - v_y n_y/c & -v_y n_z/c \\ -v_z n_x/c & -v_z n_y/c & 1 - v_z n_z/c \end{pmatrix},$$

where

$$\mathbf{n} = \frac{\mathbf{x}-\mathbf{x}_0'}{|\mathbf{x}-\mathbf{x}_0'|},$$

and $\mathbf{v}$ is the velocity of the particle at the retarted time $t' = t - |\mathbf{x}-\mathbf{x}_0'|/c$.

The determinant of the Jacobian can be easily calculated, with

$$\det[J\mathbf{f}(\mathbf{x}_0')] = 1 - \frac{1}{c}\left(v_x n_x + v_y n_y + v_z n_z\right) = 1 - \frac{\mathbf{v}\cdot\mathbf{n}}{c}.$$

In fact, there is a very general result, stating that

$$\det(\delta_{ij} - a_i b_j) = 1 - \sum_i a_i b_i.$$

Let $\kappa = 1 - \mathbf{v}\cdot\mathbf{n}/c$, then

$$\int \delta\left[\mathbf{x}' - \mathbf{r}\left(t - \frac{|\mathbf{x}-\mathbf{x}'|}{c}\right)\right] d^3x' = \int \frac{\delta(\mathbf{x}'-\mathbf{x}_0')}{|\det[J\mathbf{f}(\mathbf{x}_0')]|} d^3x' = \frac{1}{\kappa}.$$

(b) From Eq. (6.55), the electric field is

$$\mathbf{E}(\mathbf{x},t) = \frac{1}{4\pi\epsilon_0}\int \left\{ \frac{\hat{\mathbf{R}}}{R^2}\left[\rho(\mathbf{x}',t')\right]_{\text{ret}} + \frac{\hat{\mathbf{R}}}{cR}\left[\frac{\partial\rho(\mathbf{x}',t')}{\partial t'}\right]_{\text{ret}} - \frac{1}{c^2R}\left[\frac{\partial\mathbf{J}(\mathbf{x}',t')}{\partial t'}\right]_{\text{ret}} \right\} d^3x',$$

where $\hat{\mathbf{R}}$ is $\mathbf{n}$ from part (a) and $R = |\mathbf{x}-\mathbf{x}_0'|$. As pointed out by Eq. (6.57),

$$\left[\frac{\partial f(x',t')}{\partial t'}\right]_{\text{ret}} = \frac{\partial}{\partial t}\left[f(x',t')\right]_{\text{ret}},$$

and also notice that R does not depend on t, we can write the expression for electric field as

$$\mathbf{E}(\mathbf{x},t) = \frac{1}{4\pi\epsilon_0}\int \left\{ \left[\frac{\hat{\mathbf{R}}}{R^2}\rho(\mathbf{x}',t')\right]_{\text{ret}} + \frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{R}\rho(\mathbf{x}',t')\right]_{\text{ret}} - \frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{\mathbf{J}(\mathbf{x}',t')}{R}\right]_{\text{ret}} \right\} d^3x'.$$

Since $\rho(\mathbf{x}', t') = q\delta[\mathbf{x}' - \mathbf{r}(t_{\rm ret})]$ and $\mathbf{J}(\mathbf{x}', t') = \rho(\mathbf{x}', t')\mathbf{v}(t')$, following the result from part (a), we can directly write down the electri field as

$$\mathbf{E}(\mathbf{x}, t) = \frac{q}{4\pi\epsilon_0}\left\{\left[\frac{\hat{\mathbf{R}}}{\kappa R^2}\right]_{\rm ret} + \frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{\kappa R}\right]_{\rm ret} - \frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}}{\kappa R}\right]_{\rm ret}\right\}.$$

Similarly, for the magnetic field,

$$\begin{aligned}\mathbf{B}(\mathbf{x}, t) &= \frac{\mu_0}{4\pi}\int\left\{\left[\rho\mathbf{v}\times\frac{\hat{\mathbf{R}}}{R^2}\right]_{\rm ret} + \frac{1}{c}\frac{\partial}{\partial t}\left[\rho\mathbf{v}\times\frac{\hat{\mathbf{R}}}{R}\right]_{\rm ret}\right\}d^3x' \\ &= \frac{\mu_0 q}{4\pi}\left\{\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa R^2}\right]_{\rm ret} + \frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa R}\right]_{\rm ret}\right\}.\end{aligned}$$

(c) Let us first establish a few useful relations. Denote $\mathbf{R} = \mathbf{x} - \mathbf{x}'(t')$, then

$$\frac{\partial}{\partial t'}R = -\mathbf{v}(t')\cdot\hat{\mathbf{R}},$$

with

$$\mathbf{v} = \frac{d\mathbf{x}'(t')}{dt'}, \quad \hat{\mathbf{R}} = \frac{\mathbf{x} - \mathbf{x}'(t')}{|\mathbf{x} - \mathbf{x}'(t')|}.$$

Applying the chain rule, we have

$$\frac{\partial}{\partial t'}\left(\frac{1}{R}\right) = \frac{\mathbf{v}(t')\cdot\hat{\mathbf{R}}}{R^2},$$

and

$$\frac{\partial}{\partial t'}\hat{\mathbf{R}} = \frac{(\mathbf{v}(t')\cdot\hat{\mathbf{R}})\hat{\mathbf{R}} - \mathbf{v}(t')}{R}.$$

Also, for the retarted time,

$$t' = t - \frac{|\mathbf{x} - \mathbf{x}'(t')|}{c},$$

we have

$$\frac{dt}{dt'} = 1 - \frac{\mathbf{v}(t')\cdot\hat{\mathbf{R}}}{c} = \kappa.$$

Now, consider the Jefimenko expression for the magnetic field. The second term can be written as

$$\frac{\partial}{\partial t}\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa R}\right]_{\rm ret} = \frac{1}{[R]_{\rm ret}}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa}\right]_{\rm ret} + \left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa}\right]_{\rm ret}\frac{\partial}{\partial t}\left[\frac{1}{R}\right]_{\rm ret}.$$

Since

$$\frac{\partial}{\partial t}\left[\frac{1}{R}\right]_{\rm ret} = \frac{\partial}{\partial t'}\left[\frac{1}{R}\right]_{\rm ret}\frac{dt'}{dt} = \frac{1}{\kappa}\frac{\partial}{\partial t'}\left(\frac{1}{R(t')}\right) = \frac{\mathbf{v}(t')\cdot\hat{\mathbf{R}}}{\kappa R(t')^2} = \left[\frac{\mathbf{v}\cdot\hat{\mathbf{R}}}{\kappa R^2}\right]_{\rm ret},$$

then

$$\mathbf{B}(\mathbf{x}, t) = \frac{\mu_0 q}{4\pi}\left\{\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa R^2}\right]_{\rm ret} + \frac{1}{c[R]_{\rm ret}}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa}\right]_{\rm ret} + \frac{1}{c}\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa}\frac{\mathbf{v}\cdot\hat{\mathbf{R}}}{\kappa R^2}\right]_{\rm ret}\right\}$$

$$
\begin{aligned}
&= \frac{\mu_0 q}{4\pi}\left\{\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa R^2}\right]_{\rm ret}\left[1+\frac{\mathbf{v}\cdot\hat{\mathbf{R}}}{\kappa c}\right]_{\rm ret}+\frac{1}{c[R]_{\rm ret}}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa}\right]_{\rm ret}\right\} \\
&= \frac{\mu_0 q}{4\pi}\left\{\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa^2 R^2}\right]_{\rm ret}+\frac{1}{c[R]_{\rm ret}}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}\times\hat{\mathbf{R}}}{\kappa}\right]_{\rm ret}\right\},
\end{aligned}
$$

where have used

$$1+\frac{\mathbf{v}\cdot\hat{\mathbf{R}}}{\kappa c}=\frac{1}{\kappa}.$$

To verify the equivalence for the electric field, let us start with the Feynman expression. Notice that

$$
\begin{aligned}
\frac{1}{c^2}\frac{\partial^2}{\partial t^2}[\hat{\mathbf{R}}]_{\rm ret} &= \frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{1}{\kappa}\frac{\partial\hat{\mathbf{R}}}{\partial t'}\right]_{\rm ret} \\
&= \frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{(\mathbf{v}\cdot\hat{\mathbf{R}})\hat{\mathbf{R}}-\mathbf{v}}{\kappa R}\right]_{\rm ret} \\
&= \frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{(\mathbf{v}\cdot\hat{\mathbf{R}})\hat{\mathbf{R}}}{\kappa R}\right]_{\rm ret}-\frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}}{\kappa R}\right]_{\rm ret} \\
&= \frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{\kappa R}(1-\kappa)\right]_{\rm ret}-\frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}}{\kappa R}\right]_{\rm ret} \\
&= \frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{\kappa R}\right]_{\rm ret}-\frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{R}\right]_{\rm ret}-\frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}}{\kappa R}\right]_{\rm ret},
\end{aligned}
$$

and

$$
\begin{aligned}
\frac{[R]_{\rm ret}}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{R^2}\right]_{\rm ret} &= \frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{R}\right]_{\rm ret}-\frac{1}{c}\left[\frac{\hat{\mathbf{R}}}{R^2}\right]_{\rm ret}\frac{\partial}{\partial t}[R]_{\rm ret} \\
&= \frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{R}\right]_{\rm ret}+\frac{\mathbf{v}\cdot\hat{\mathbf{R}}}{\kappa c}\left[\frac{\hat{\mathbf{R}}}{R^2}\right]_{\rm ret},
\end{aligned}
$$

the Feynman expression now becomes

$$
\begin{aligned}
\mathbf{E}(\mathbf{x},t) &= \frac{q}{4\pi\epsilon_0}\left\{\left[\frac{\hat{\mathbf{R}}}{R^2}\right]_{\rm ret}+\frac{[R]_{\rm ret}}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{R^2}\right]_{\rm ret}+\frac{1}{c^2}\frac{\partial^2}{\partial t^2}[\hat{\mathbf{R}}]_{\rm ret}\right\} \\
&= \frac{q}{4\pi\epsilon_0}\left\{\left[\frac{\hat{\mathbf{R}}}{R^2}\right]_{\rm ret}+\frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{R}\right]_{\rm ret}+\frac{\mathbf{v}\cdot\hat{\mathbf{R}}}{\kappa c}\left[\frac{\hat{\mathbf{R}}}{R^2}\right]_{\rm ret}\right. \\
&\qquad\left.+\frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{\kappa R}\right]_{\rm ret}-\frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{R}\right]_{\rm ret}-\frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}}{\kappa R}\right]_{\rm ret}\right\} \\
&= \frac{q}{4\pi\epsilon_0}\left\{\left[\frac{\hat{\mathbf{R}}}{R^2}\right]_{\rm ret}\left[1+\frac{\mathbf{v}\cdot\hat{\mathbf{R}}}{\kappa c}\right]_{\rm ret}+\frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{\kappa R}\right]_{\rm ret}-\frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}}{\kappa R}\right]_{\rm ret}\right\} \\
&= \frac{q}{4\pi\epsilon_0}\left\{\left[\frac{\hat{\mathbf{R}}}{\kappa R^2}\right]_{\rm ret}+\frac{1}{c}\frac{\partial}{\partial t}\left[\frac{\hat{\mathbf{R}}}{\kappa R}\right]_{\rm ret}-\frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{\mathbf{v}}{\kappa R}\right]_{\rm ret}\right\},
\end{aligned}
$$

which is the Jefimenko expression.